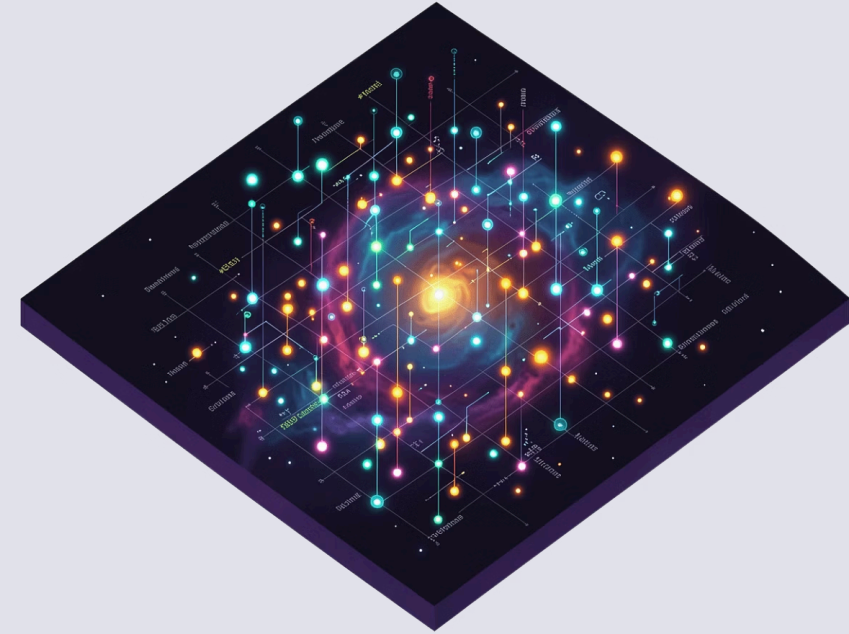


DBSCAN: Density-Based Clustering Made Simple

An introduction to unsupervised density-based machine learning for engineering students

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What is DBSCAN?

Density-Based Spatial Clustering of Applications with Noise — an unsupervised ML algorithm that groups tightly packed points and marks isolated ones as **outliers (noise)**.

🔍 Auto-discovers clusters

No need to specify how many clusters exist — it figures it out!

♦ Any shape

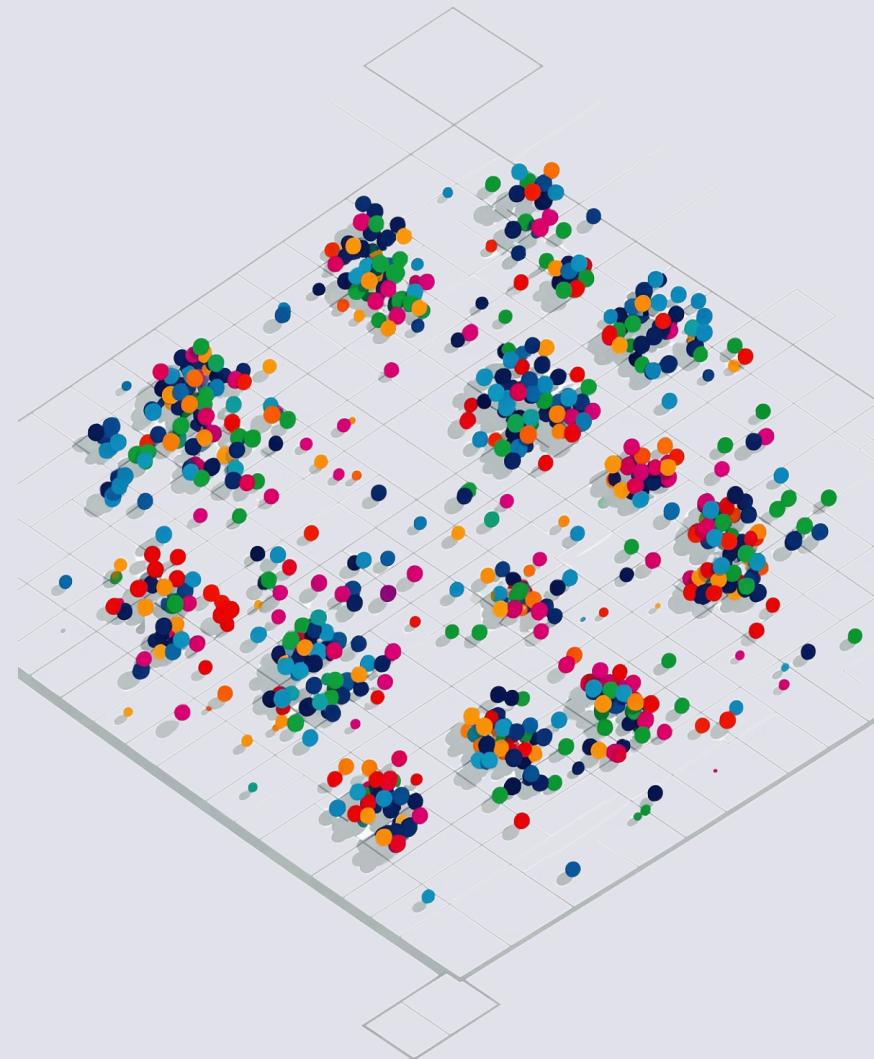
Finds clusters of any shape — not just round blobs

🚫 Handles outliers

Naturally labels isolated points as **noise**

🏠 Real-world ready

Anomaly detection, spatial analysis, image segmentation

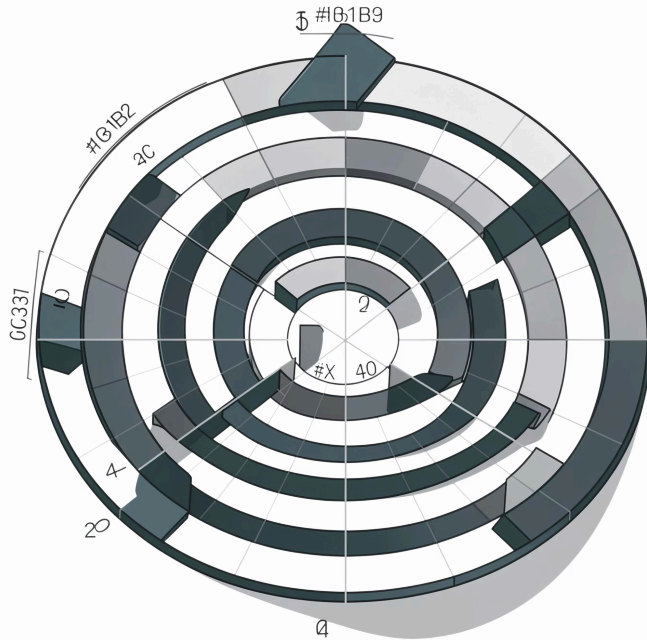


Why Not Just Use K-Means?

K-Means forces you to decide the number of clusters upfront and struggles with odd shapes. DBSCAN is smarter — it finds groups naturally and labels loners as "noise."

Feature	K-Means	DBSCAN
Number of clusters	Must specify upfront	Discovered automatically ✓
Cluster shape	Only round/spherical	Any arbitrary shape ✓
Outlier handling	Forced into a cluster	Labelled as noise ✓
Noise robustness	Poor ✗	Excellent ✓
Works with spatial data	Limited ✗	Excellent ✓

DBSCAN's 3 Magic Ingredients



Two Parameters You Control

Three Point Types

- **Core Point:** $\geq \text{MinPts}$ neighbours within $\epsilon \rightarrow$ heart of a cluster
- **Border Point:** Fewer neighbours, but within ϵ of a Core Point
- **Noise Point (-1):** Isolated — belongs to no cluster $\rightarrow$ outlier

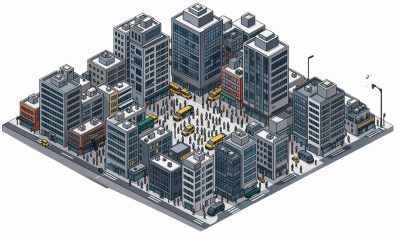
ϵ (Epsilon) — "Friendship Radius"

How close must two points be to be neighbours? **e.g. eps = 5**

MinPts — Minimum Neighbours

Minimum neighbours inside ϵ to form a dense region. **e.g. min_samples = 5**

Core, Border & Noise — Visualised



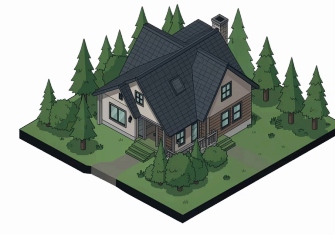
 Core Point

The **busy city centre** — dense, many neighbours within ϵ . Forms the heart of every cluster.



 Border Point

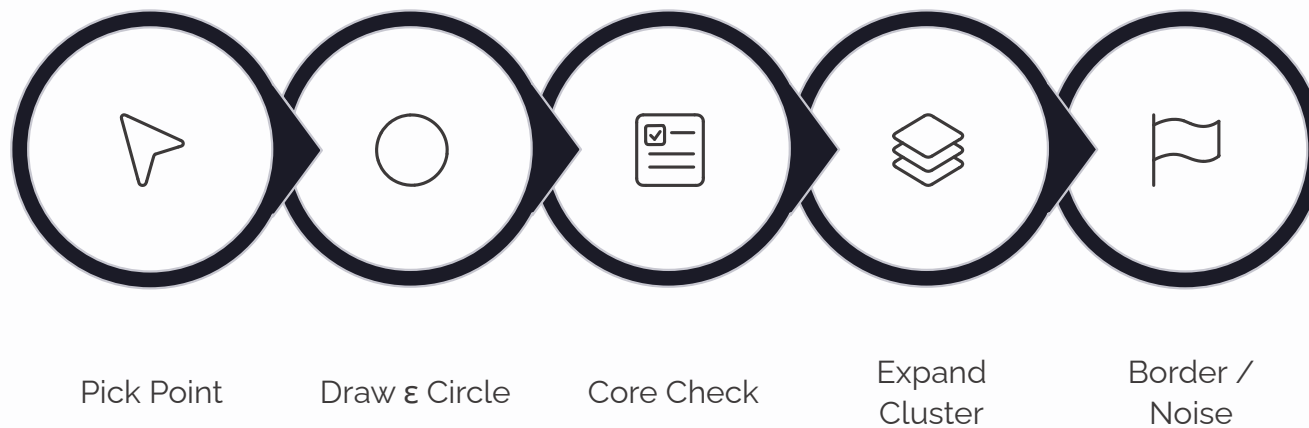
The **suburbs** — fewer neighbours, but reachable from the city centre. Part of the cluster, on the edge.



 Noise Point (-1)

An **isolated house in the middle of nowhere** — not close enough to any cluster. Labelled **-1** .

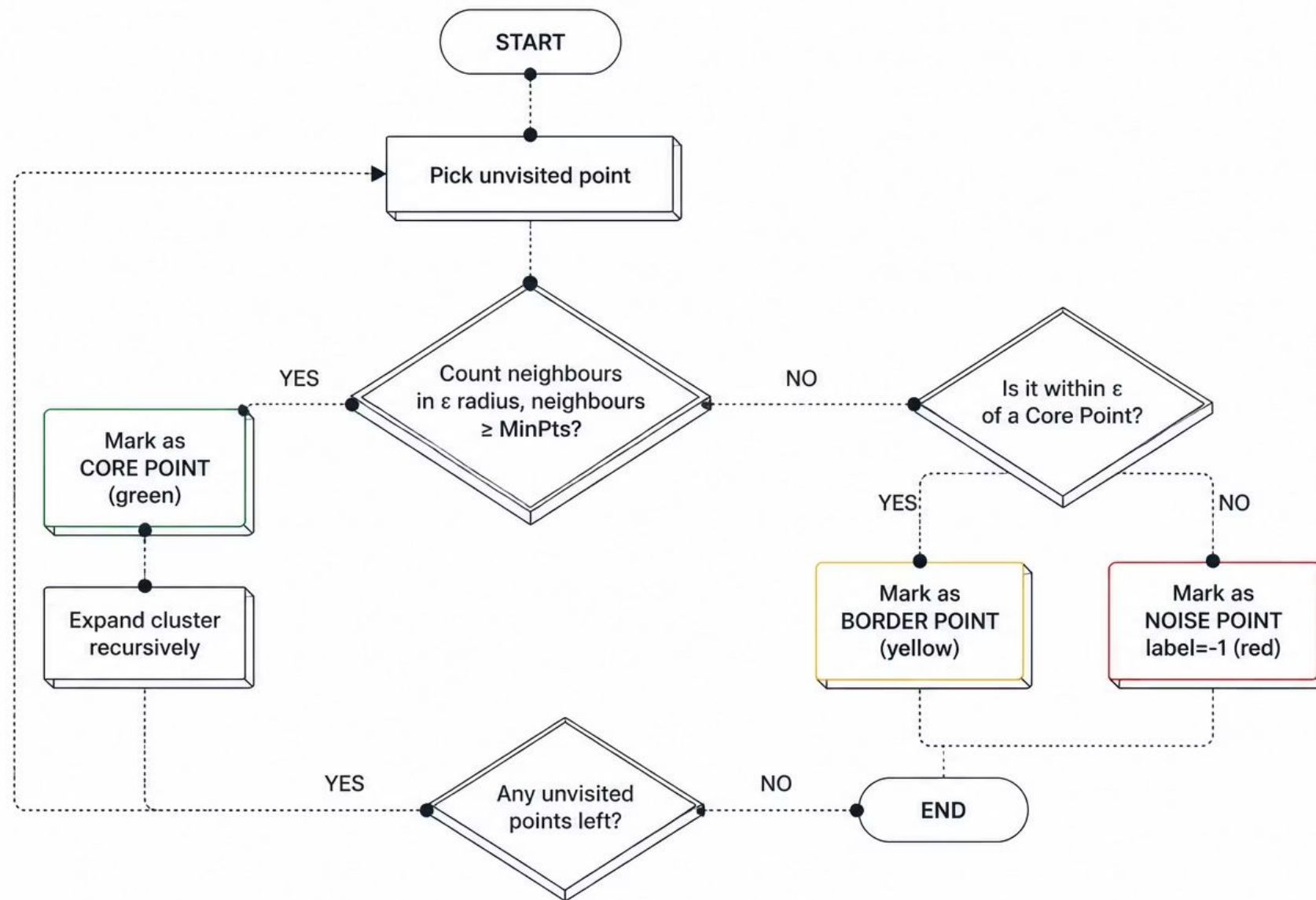
DBSCAN Algorithm — Step by Step



- 1 Pick any unvisited point
- 2 Draw ϵ circle $\rightarrow$ count neighbours
- 3 $\geq \text{MinPts?}$ $\rightarrow$ Core Point $\rightarrow$ new cluster
- 4 Expand cluster recursively
- 5 Near core but few neighbours $\rightarrow$ Border
- 6 Completely isolated $\rightarrow$ Noise (-1)
- 7 Repeat until every point is visited

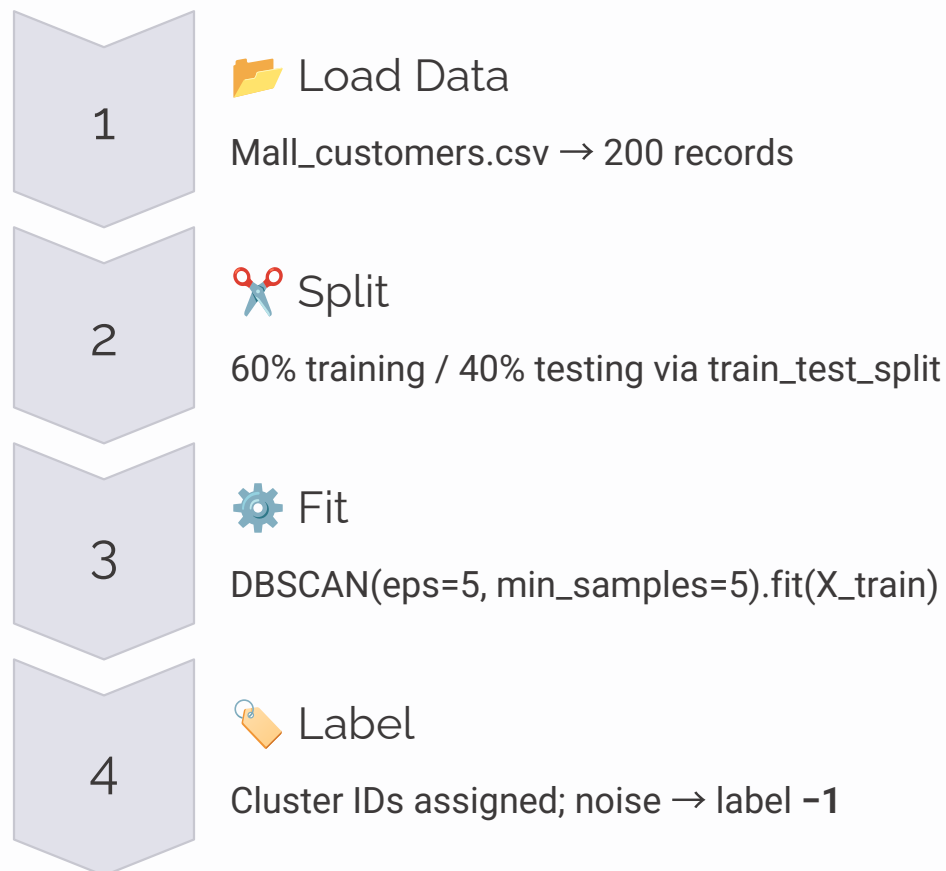
DBSCAN Flowchart

Follow this flowchart to see exactly how DBSCAN makes a decision for every data point:



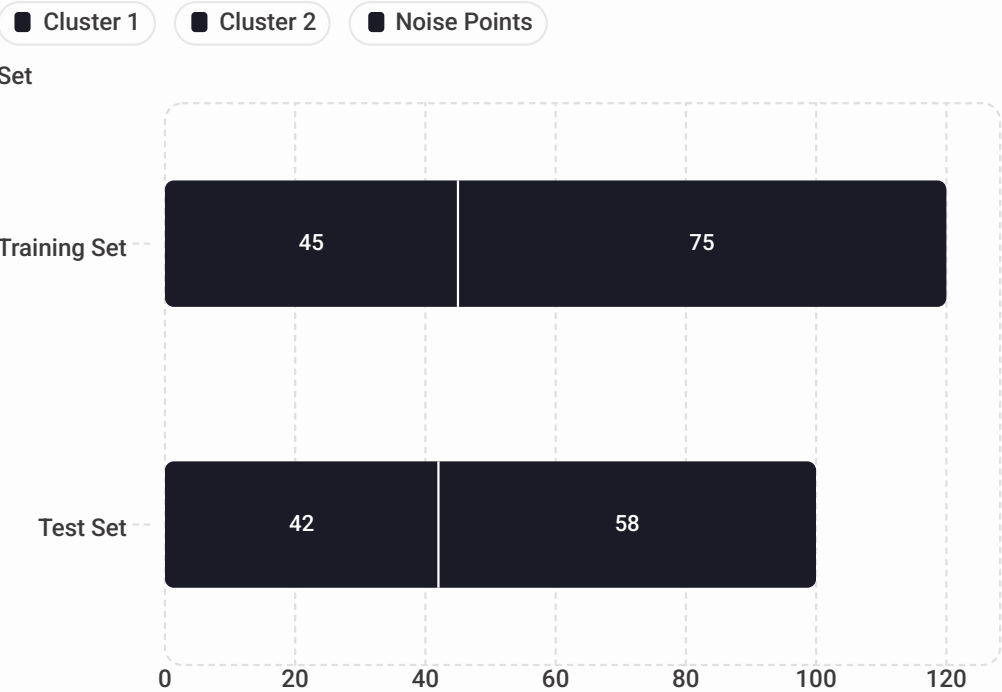
Python Implementation — Mall Customers Dataset

Using **Annual Income (k\$)** and **Spending Score (1–100)** as features, with **eps=5** and **min_samples=5** on 200 customer records.



i DBSCAN is a fitting model, not predictive — the test set is clustered independently, not predicted from training labels.

Results: What Did DBSCAN Find?



Key Findings

Cluster 1

Annual Income $\approx$ 60k\$ |
Spending Score $\approx$ 60
**High earners, high
spenders**

Cluster 2

Annual Income $\approx$ 60k\$ |
Spending Score $\approx$ 40
**High earners, moderate
spenders**

Noise Points

75 in training, 58 in test
— spread across feature
space

Stable Algorithm



Consistent clustering
across train & test
confirms algorithm
stability

2

Clusters Found

Consistently in both training and test sets

75

Training Noise Points

Isolated customers – labelled -1

58

Test Noise Points

Outliers in the test set

200

Customer Records

Total Mall_customers.csv dataset size

DBSCAN automatically discovered structure in raw customer data – no cluster count required.

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Industrial Use Cases of DBSCAN



Predictive Maintenance

Sensor data clusters reveal machine health — noise = anomaly alert **before breakdown**. DBSCAN is the best algorithm for monitoring the time evolution of industrial machine health status.



Robotic Welding QC

Real-time current/voltage clusters flag **defective welds instantly** using hierarchical DBSCAN in robotic welding cells.



Fraud Detection

Transactions outside all clusters = **suspicious activity flagged automatically**



Smart Traffic

GPS data clustered to detect **congestion hotspots** in real time



Healthcare

Map disease outbreak clusters geographically — **no cluster count needed**

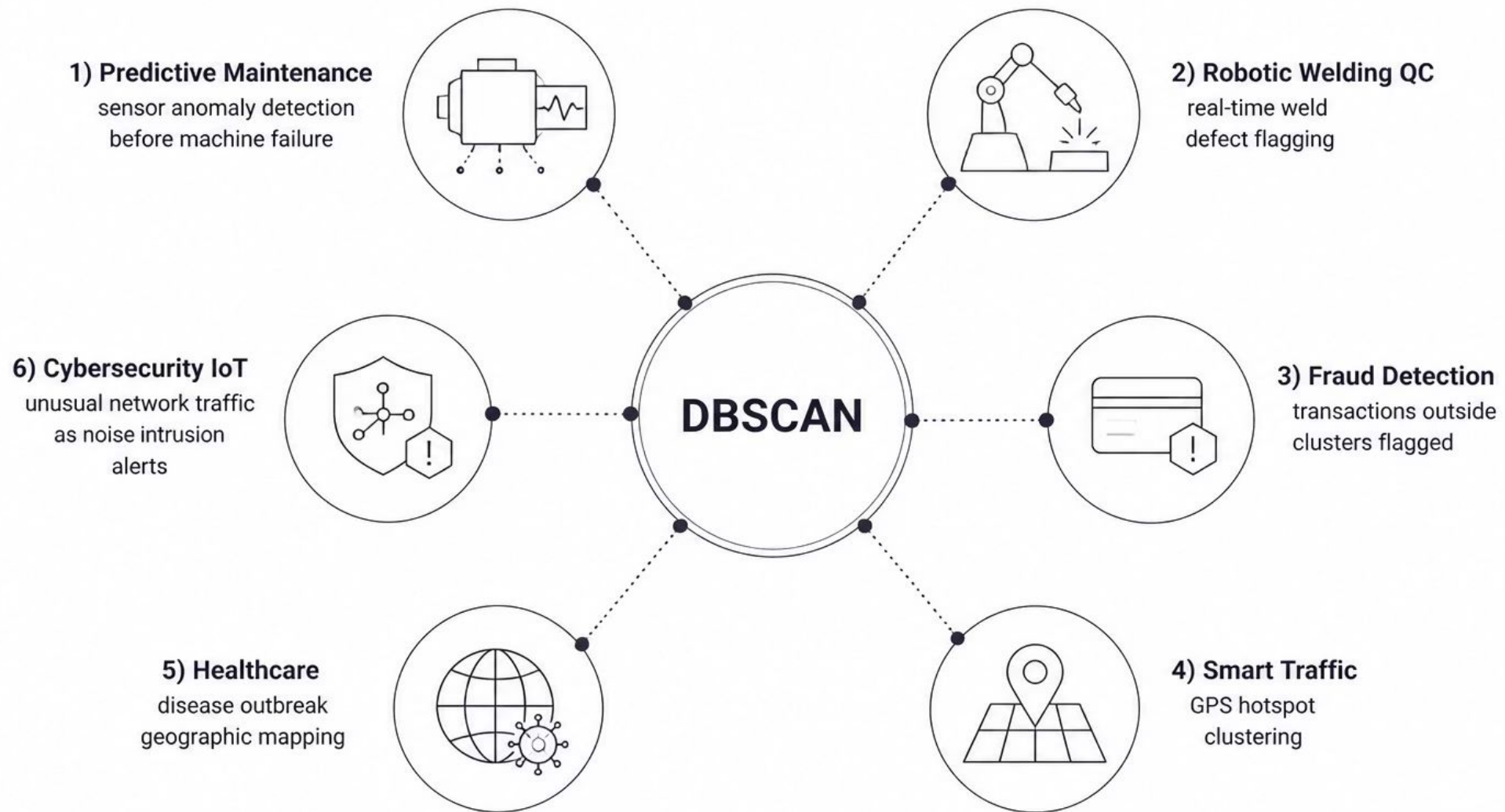


Cybersecurity / IoT

Unusual network traffic = noise = **intrusion alert**

- No cluster count needed
 - DBSCAN discovers the number of clusters automatically from data density
- Two simple parameters
 - ϵ (epsilon) and MinPts are all you need to configure
- Noise is information
 - Points labelled -1 are outliers – valuable for anomaly detection
- Any shape, any domain
 - Works on spatial, sensor, financial, and healthcare data alike

Industrial Use Cases — At a Glance



Sources & References

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On Sensor Data Clustering for Machine Status Monitoring and Its Application to Predictive Maintenance

Ferrari et al. – SENSORS Journal 2023. Demonstrates DBSCAN as the best algorithm for clustering industrial sensor data to monitor machine health over time.

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MDPI Information Journal 2025. Application of hierarchical DBSCAN for real-time weld quality monitoring in robotic manufacturing cells.